DOES MEMORY PROVENANCE MATTER? MATCHED SUCCESS–FAILURE SWAPS LEAVE THE CAUSAL SIGN UNRESOLVED

Anonymous authors
Paper under double-blind review

ABSTRACT

Self-evolving agents convert previous trajectories into reusable textual memory. Existing analyses usually treat the retrieved text as the relevant state variable. We study a second variable carried into that state: the provenance of the trajectory from which the memory was written. A recent financial-agent audit reports a sharp ReasoningBank split. Success-derived retrievals achieve 0.931 accuracy over 101 cases, while failure-derived retrievals achieve 0.647 over 34 cases. All ten reported persistent W$\rightarrow$W failures occur in the failure-derived stratum. We formalize provenance as an upstream variable in memory construction and execute two controlled matched-memory endpoints on released WebArena evidence. The first endpoint evaluates terminal success over four pre-outcome information-equivalent pairs and 48 completed requests. Its success-derived minus failure-derived effect is +0.1667, while the frozen 100,000-permutation test gives $p = 0.0785$ and leaves the support gate closed. With four independent units, the exact sign-flip test cannot attain a one-sided value below $1/16 = 0.0625$, so this endpoint is structurally non-decisive under the frozen 0.05 rule. The second endpoint uses a disjoint cohort. Twenty-three of 24 frozen candidates pass the information-equivalence gate before action outcomes, yielding 276 completed requests with zero support failures. Success-derived memory changes reference-action match by -0.0942 relative to failure-derived memory. The counterevidence-side permutation value is $p = 0.0792$, so that gate also remains closed. Paired next actions differ in 21.0% of seeds. This second endpoint localizes action selection and is not a terminal-quality surrogate. The frozen causal hypothesis therefore remains active without supported or decisive counterevidence. These results separate a strong provenance-conditioned association from its unresolved causal sign and make provenance an explicit audit variable for future memory trust decisions.

1 INTRODUCTION

Self-evolving agents increasingly learn through external memory. A successful episode can become a reusable workflow. A failed episode can become a corrective reflection. Both objects may later be retrieved for a new task. Systems such as ReasoningBank make this loop explicit by converting trajectories into reasoning memories that influence future behavior (Ouyang et al., 2026). Related systems store workflows or verbal feedback as persistent state (Wang et al., 2024; Shinn et al., 2023).

The usual abstraction treats memory content as the object that matters. Content-centric systems such as A-MEM and MemRL already improve memory organization, retrieval, or runtime utility (Xu et al., 2025; Zhang et al., 2026). A second literature makes provenance explicit: a recent survey organizes evidence tracing and execution provenance for LLM agents (Wang et al., 2026), while Memory Provenance Laundering shows that consolidation can preserve an action trigger after low-trust source provenance is obscured and proposes a provenance firewall (Xu et al., 2026). These works establish provenance as a memory-governance object, so we do not claim memory provenance itself is new. Our narrower variable is *process provenance*: whether semantically matched guidance was derived from a successful or failed trajectory. ReasoningBank exposes this intervention because successful and failed trajectories are summarized under different reflection objectives even when the resulting procedural advice can overlap.

Crucially, failure provenance has no justified negative sign a priori. Learning from Failure converts failed computer-use trajectories into diagnoses and patches that improve later behavior (Sun et al., 2026), and controlled multi-round skill-evolution experiments find useful skills in conditions that include failure feedback (Liu et al., 2026). These results rule out the shortcut “failure-derived means harmful.” The missing scientific object is therefore a same-information provenance intervention that can determine whether origin has an effect at all, and if so with what sign.

A recent audit of self-evolving financial agents provides a strong empirical signal for this distinction (Li & Zhu, 2026). In its ReasoningBank analysis, success-derived retrievals have reported accuracy 0.931 over 101 cases. Failure-derived retrievals have reported accuracy 0.647 over 34 cases. The audit further reports ten persistent W→W failures, where a task fails before evolution and continues to fail afterward. Every one of these persistent failures appears in the failure-derived retrieval stratum. Figure 2 summarizes the released counts.

This association points to a causal question with direct implications for memory governance: does provenance itself change future behavior after semantic guidance is controlled? Task difficulty provides the strongest competing explanation. Hard tasks are more likely to fail during acquisition. Those same tasks can later retrieve memories derived from failure. A descriptive provenance gap can therefore emerge even when provenance has no causal effect. We test this identification problem with matched provenance interventions.

The first intervention freezes future queries before memory writing and admits only pairs that pass a pre-outcome information-equivalence gate. Four pairs qualify. Across 48 terminal requests, success-derived memory increases mean correctness by 0.1667. The preregistered one-sided permutation value is $p = 0.0785$, so the support gate remains closed. We then execute the already specified early-action endpoint on a disjoint cohort. Twenty-three information-equivalent pairs produce 276 completed action requests. The success-derived minus failure-derived reference-action match difference is -0.0942. Its counterevidence-side permutation value is $p = 0.0792$, leaving that gate closed as well. The two controlled endpoints therefore disagree in direction under their frozen decision rules.

The supported empirical findings are deliberately narrower than the causal hypothesis. C1 and C2 are source-bound observations from the financial audit: the provenance-conditioned accuracy gap is large, and every reported persistent W→W case lies in the failure-derived stratum. C3 is a mechanism fact from the ReasoningBank writer contract: trajectory outcome selects the reflection mode that produces persistent memory. C4 remains an active causal hypothesis. C5 remains a design implication whose downstream governance benefit is not claimed. We execute controlled endpoints to constrain C4 without promoting either endpoint beyond its frozen decision rule. This establishes a concrete confirmation burden for provenance-aware memory governance.

Contribution and closest-work boundary. The source-bound financial association motivates the question; it is not an independent novelty claim. Our contribution is a process-provenance identification object: (i) define the treatment as success-versus-failure trajectory origin after actionable guidance has passed a pre-outcome information-equivalence gate, (ii) execute matched same-information swaps with an explicit small-sample power boundary, and (iii) adjudicate across disjoint terminal and early-action endpoints without forcing a direction when neither preregistered gate passes. Existing systems already track provenance, authorize outcome-conditioned memory mutation, and show that success- and failure-derived experience can have different task-dependent value. We therefore do not claim provenance-aware memory, outcome-aware memory governance, or harmful failure memory as new; the present result is precisely that a stable causal sign remains unresolved under the controlled swaps.

2 Memory Provenance as Persistent Agent State

Consider an acquisition episode with task x and trajectory τ. Its terminal outcome is $z \in \{0, 1\}$. A memory writer receives the episode and produces a textual memory

$$M = G(\tau, z). \tag{1}$$

ReasoningBank instantiates G with outcome-conditioned reflection. The successful branch asks the writer to explain the strategy that worked. The failure branch asks the writer to identify the cause of

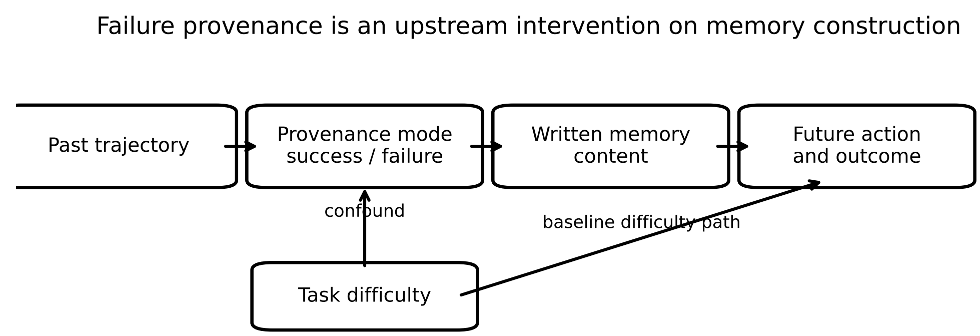

Figure 1: Causal structure of provenance-conditioned memory. Task difficulty affects which trajectories fail and also affects future outcomes. The matched-memory intervention blocks this confound by fixing the future query and matching semantic guidance across provenance conditions.

failure and construct an improved approach. We call the branch variable $P \in \{S, F\}$ the memory provenance.

A future query x' retrieves a memory through a similarity function $R(x', x)$. The retrieval key can be identical even when the written memory differs because the key is the historical task description. Future behavior then follows

$$Y = H(x', M(P), S), \tag{2}$$

where S denotes the rest of the agent state. Provenance reaches future behavior through memory content. It is therefore an upstream causal variable with direct influence on the persistent state.

2.1 Target estimand

Let $C(M)$ denote the semantic guidance contained in a memory. The quantity of interest is the effect of provenance under matched guidance:

$$\Delta_P(x') = \mathbb{E}[Y \mid do(P = F), C(M) = c, x'] - \mathbb{E}[Y \mid do(P = S), C(M) = c, x']. \tag{3}$$

For a success indicator, a negative Δ_P means failure provenance reduces future success under matched information. For an error indicator, the sign reverses.

Task difficulty D creates the central identification challenge. Difficult acquisition tasks yield more failed trajectories. They also remain difficult when encountered later. Conditioning only on observed provenance therefore estimates a mixture of provenance effect and task difficulty. Equation 3 motivates a direct intervention on the retrieved memory while the future task remains fixed.

2.2 Matched provenance pairs

A provenance pair contains two memories with the same intended procedural guidance. One member is derived from a successful trajectory. The other member is derived from a failed trajectory. Matching is performed over the task family and the actionable recommendation. Each pair is checked for information asymmetry before it enters the experiment. A pair is rejected when one memory contains an extra fact that can independently solve the future task.

This construction turns provenance into an experimentally manipulable state variable. It also exposes a design principle for memory systems: a textual memory is incomplete as an audit object when its learning process has been discarded. Provenance metadata preserves evidence about whether the remembered strategy emerged from demonstrated success or from a corrective interpretation of failure.

Table 1: Provenance-conditioned ReasoningBank outcomes from the released financial-agent audit.

Retrieved memory provenance	Cases	Correct	Accuracy
Success-derived	101	94	0.931
Failure-derived	34	22	0.647

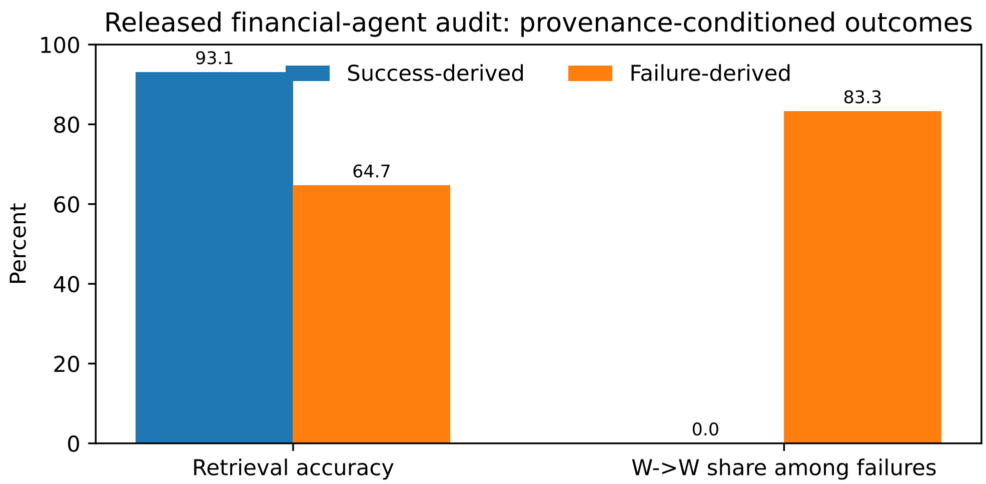

Figure 2: The released audit shows a large accuracy gap by memory provenance. Persistent W→W failures are concentrated in failure-derived retrievals.

3 Released Evidence for a Provenance Effect

We begin from the public aggregate audit of ReasoningBank on benign AgentDojo Banking endpoints (Li & Zhu, 2026). The audit separates retrievals according to whether the retrieved memory was derived from a successful episode or from a failed episode. Table 1 reproduces the reported counts after converting the rounded accuracies back to their integer support.

The observed accuracy gap is 28.4 percentage points. This is a large effect at the retrieval stratum level. The transition audit gives the provenance signal a sharper interpretation. ReasoningBank records 25 W→C recoveries and 9 C→W regressions. Ten W→W persistent failures remain after evolution. All ten belong to the failure-derived retrieval stratum.

The rounded provenance accuracies imply 12 failures among 34 failure-derived retrievals and 7 failures among 101 success-derived retrievals. The ten W→W cases therefore account for roughly 83.3% of failures in the failure-derived stratum. The success-derived stratum contains no W→W case in the audit. Figure 2 visualizes this decomposition.

3.1 The signal is strongest for persistence

The transition structure changes the scientific interpretation of the aggregate accuracy gap. Failure-derived memory is tightly associated with errors that persist across self-evolution. The source audit identifies task difficulty as a confound for this association. Our causal formulation preserves the persistence signal as the main target and removes task difficulty through an intervention on memory provenance.

A simple arithmetic check also constrains the mechanism. The failure-derived stratum contains roughly two failures beyond the ten W→W cases. The success-derived stratum contains seven failures. The provenance signal therefore concentrates on persistent error much more strongly than on newly induced regression. This pattern motivates a paper claim centered on failure persistence.

Table 2: Terminal-success matched-memory swap. Δ is success-derived minus failure-derived terminal success.

Released task	Success-derived	Failure-derived	Δ
167	0.000	0.000	0.000
23	1.000	0.667	+0.333
387	0.000	0.000	0.000
388	0.500	0.167	+0.333
Mean	0.375	0.208	+0.167

3.2 Independent mechanism evidence

Memory Reward Inflation studies a different memory architecture and identifies an amplification channel for erroneous memories (Asadolahi et al., 2026). Incorrect memories can receive inflated utility scores, which increases their future reuse. The paper also shows that this behavior can persist under similarity-based retrieval. This mechanism is compatible with the provenance effect studied here. A failure-derived memory can carry corrective advice while remaining anchored to a trajectory whose process already demonstrated an error mode. Repeated reuse can then sustain that mode when the corrective abstraction is incomplete.

Together, the financial-agent audit and reward-inflation results motivate a causal model in which memory provenance can change the distribution of future behavior. Section 4 tests the directional provenance hypothesis with two frozen controlled endpoints.

4 Controlled Provenance Intervention

We execute matched provenance interventions on released WebArena evidence using the first-party ReasoningBank memory-writing contract. The financial AgentDojo audit supplies the motivating association. The controlled bridge isolates the memory channel on a public execution substrate.

4.1 Terminal-success endpoint

We freeze six candidate future tasks before memory writing or downstream evaluation. For each task, the writer receives paired acquisition traces under its success-reflection objective and its failure-reflection objective. The downstream query evidence remains fixed within each pair.

A pair enters terminal evaluation only after an information-equivalence gate. Both memories must parse as valid ReasoningBank items. Their embedding cosine must be at least 0.75. An independent verifier must judge the actionable guidance equivalent and find no task-solving fact unique to either memory. Literal benchmark-reference leakage excludes a pair before downstream outcomes. Four of the six frozen candidates pass. Their cosine values range from 0.902 to 0.931.

The evidence packets contain repeated global browser chrome. A condition-blind projection removes complete menu blocks before the clean run. It also drops blank lines and exact duplicate lines. The projection is frozen without consulting downstream answers. A context preflight shows that the largest prompt occupies 5,209 input tokens inside a 12,288-token allocation. Qwen2.5-32B remains the executor at temperature 0.6. The pair set and seed schedule remain frozen.

Each qualified pair receives six paired seeds per provenance condition. Terminal correctness uses the WebArena must-include evaluator. Table 2 reports pair-level success rates.

The mean controlled difference is +0.1667. The preregistered support gate requires a difference of at least 0.15 together with a one-sided stratified permutation value below 0.05. The magnitude criterion is met. The 100,000-permutation test gives $p = 0.0785$, so the support gate remains closed. More fundamentally, four independent task units permit only $2^4 = 16$ exact sign patterns; the smallest attainable exact one-sided value is therefore $1/16 = 0.0625$. R4 is structurally incapable of clearing the frozen 0.05 gate and is treated as directional, non-decisive evidence. The counterevidence gate remains closed as well.

Table 3: Two frozen controlled endpoints. Support requires $\Delta \geq 0.15$ with $p_S < 0.05$. Counterevidence requires $\Delta \leq -0.15$ with $p_F < 0.05$.

Endpoint	Units	Requests	Δ	Relevant p	Gate
Terminal success	4	48	+0.167	$p_S = 0.0785$	closed
Reference-action match	23	276	−0.094	$p_F = 0.0792$	closed

4.2 Disjoint early-action endpoint

The manuscript specifies action divergence before terminal feedback as a second endpoint. We execute that endpoint on a disjoint cohort after the terminal run. Candidate selection occurs before counterfactual memory writing. A released successful trajectory supplies a frozen reference action at an intermediate browser state. This reference is external to the provenance intervention.

The candidate rule selects the earliest visible single-click decision at step two or later. Prior D2 task IDs are excluded. The remaining tasks are ordered by numeric task ID. The first 24 candidates are frozen. Their compact process traces are byte-identical across writer conditions. Terminal success labels are removed from those traces before the ReasoningBank success-reflection and failure-reflection interventions.

Twenty-three of the 24 candidates pass the pre-outcome information-equivalence gate. The minimum gate is 12. One candidate is excluded because the independent verifier returns an unparsable audit response; this support failure carries no scientific verdict. All 23 eligible pairs enter downstream execution. Each receives six paired seeds under each memory condition, producing 276 completed requests with zero support failures.

The endpoint asks for the next click from the frozen browser state. The released successful trajectory provides the reference click. We use this endpoint only to localize whether provenance changes action selection before terminal feedback. Reference-action match is not treated as a surrogate for task success: a different click can still lead to a correct trajectory, so terminal quality remains the distinct endpoint above. Statistical inference uses tasks as the independent units. The preregistered test sign-flips each task-level success-derived minus failure-derived reference-match difference over 100,000 permutations.

For the early-action endpoint, success-derived memory changes reference-action match by -0.0942 relative to failure-derived memory. Its support-side value is 0.9841. The counterevidence-side value is 0.0792. Paired action choices differ in 21.0% of seeds. Error-consistent divergence in the hypothesized direction occurs in 0.7% of seeds. The reverse pattern occurs in 10.1%. These action-level quantities are mechanism-localization evidence only; they do not establish that either provenance condition has higher terminal task quality.

4.3 Cross-endpoint adjudication

Table 3 shows opposite signs across two different readouts. The R4 terminal endpoint points toward the frozen C4 hypothesis, but its four-unit test is non-decisive under the frozen significance rule. R6 measures early action selection and does not adjudicate terminal task quality. Its negative reference-match difference therefore cannot be promoted to terminal counterevidence. Neither preregistered gate passes. We keep C4 as an active unrefuted hypothesis and assign no supported causal sign to the current controlled evidence. The frozen thresholds remain unchanged.

C5 remains a design implication. Provenance is an upstream variable in the memory writer, and the controlled interventions expose behavior differences under matched query state. A downstream performance benefit from provenance-aware governance remains outside the supported claim set.

5 Related Work

Content- and utility-centric agent memory. A-MEM builds dynamically linked agentic memories (Xu et al., 2025), while MemRL learns runtime utility over episodic memories (Zhang et al., 2026). ReasoningBank, Agent Workflow Memory, and Reflexion likewise turn trajectories or feedback into persistent reusable state (Ouyang et al., 2026; Wang et al., 2024; Shinn et al., 2023). These systems establish memory construction and retrieval; our variable is not content relevance alone but the process origin of semantically matched guidance.

Provenance, outcome evidence, and trust. The evidence-tracing survey treats execution provenance, lineage, and provenance-bearing memory as trust objects (Wang et al., 2026). Memory Provenance Laundering demonstrates that consolidation can hide low-trust source provenance while preserving action triggers and proposes a non-amplification firewall (Xu et al., 2026). MutMem records signed positive/negative outcome evidence and provenance-bearing authorized memory mutations (Saidi, 2026). Thus provenance and outcome-conditioned trust metadata are already explicit memory-governance objects. Our closest-work boundary is narrower: success-derived versus failure-derived *generating-process* provenance under a same-information intervention, with no assumed sign and no new governance-policy claim.

Learning from failure. Failure-derived information can be useful. Learning from Failure converts failed computer-use trajectories into inference-time diagnoses and patches (Sun et al., 2026); Rethinking Self-Evolving Agent Skills finds useful evolved skills in feedback conditions containing failures (Liu et al., 2026). Experiential Reflective Learning directly separates success- and failure-derived heuristics and reports task-dependent advantages, including stronger failure-derived guidance on its Search split (Allard et al., 2026). Memory Reward Inflation separately shows how erroneous memories can be overvalued and propagated (Asadolahi et al., 2026). Together these works rule out a blanket "failure memory is bad" story. They do not, however, hold actionable guidance information-equivalent while changing only its generating-process origin; that residual provenance effect is what our matched swaps attempt to identify.

Causal identification and claim boundary. The financial-agent audit reports a strong success/failure provenance association (Li & Zhu, 2026), but task difficulty can confound that gap. Existing memory ablations often change both information and provenance. Our matched provenance swap holds the future query and procedural guidance fixed, then examines two frozen endpoints. Because neither endpoint passes its preregistered gate and their signs disagree, the paper's contribution is the identification object and resulting unresolved boundary—not a directional causal theorem or a validated provenance-governance policy.

6 LIMITATIONS

The financial AgentDojo audit remains the source of the 0.931-versus-0.647 provenance association and the ten persistent W→W cases. Its public release does not currently expose the per-query ReasoningBank artifacts needed for an exact matched swap. A reconstruction preflight recovered the benchmark and ReasoningBank implementation. The exact source configuration uses Qwen 3.7 Flash with qwen3.7-text-embedding, and the current execution environment lacks auditable access to that configuration. An exact financial replication therefore remains support and experiment debt. These access constraints carry no scientific authority.

The terminal controlled swap contains four information-equivalent pairs. Its +0.1667 effect exceeds the frozen magnitude floor, while the one-sided permutation value is $p = 0.0785$. Two pairs contribute +0.333 and two contribute zero. With four independent units, an exact sign-flip test has only 16 patterns and cannot attain a one-sided value below 0.0625, so R4 could not satisfy the frozen $p < 0.05$ rule regardless of the observed sign allocation. The early-action endpoint contains 23 disjoint information-equivalent pairs and 276 complete requests. Its success-derived minus failure-derived reference-match difference is -0.0942, with counterevidence-side $p = 0.0792$. Reference-action match localizes action selection and is not a validated substitute for terminal task success. Both preregistered gates remain closed. The current controlled evidence is therefore inconclusive for a stable causal sign. No threshold is changed after observing either endpoint.

Semantic equivalence remains an experimental design dependency. The terminal endpoint uses an embedding-cosine floor and an independent verifier. The early-action endpoint repeats the same structure on a larger disjoint cohort. One early-action pair is excluded because the verifier response is unparsable. That exclusion is a support failure and occurs before downstream outcomes. A future confirmation should independently calibrate equivalence reliability without reusing the present outcome set.

We also preregistered an exact-information intervention in which actionable guidance is byte-identical and only structured provenance metadata changes. Its frozen target requires ten unique review tasks. The released shopping split contains only five unique tasks under the preregistered eligibility contract, so the experiment stops before any model request. This is a support-capacity failure and has no scientific verdict. Lowering the target after the capacity audit would violate the frozen design.

The reported financial provenance accuracies are rounded to three decimals. The integer correct counts used in our decomposition are the unique counts consistent with the reported support sizes and rounded accuracies. The transition allocation beyond the ten W→W cases is therefore derived from public aggregate statistics.

The current controlled endpoints use WebArena as a public bridge, while the motivating observation comes from a financial AgentDojo setting. Replication in the motivating environment remains necessary for a source-faithful causal effect estimate. U1 therefore remains unsupported. Provenance-aware governance has also not been shown to improve downstream performance, so U2 remains unsupported.

7 Conclusion

Self-evolving memory carries process history. The released financial-agent audit shows a large accuracy gap by memory provenance and concentrates every reported persistent W→W failure in failure-derived retrievals. ReasoningBank makes provenance an upstream intervention because successful trajectories and failed trajectories are written through different reflection objectives.

We operationalize the causal question with pre-outcome information-equivalence gates and two controlled endpoints. The four-pair terminal endpoint yields +16.7 percentage points in the hypothesized direction with $p = 0.0785$, while its task count makes the frozen 0.05 gate unattainable. The disjoint 23-pair endpoint shows that provenance can alter early action selection, but reference-action match is not a terminal-quality surrogate. Neither frozen gate passes. C4 therefore remains an active unrefuted hypothesis with a substantially sharper evidence boundary.

The resulting audit separates a strong provenance-conditioned association from its unresolved causal sign. Provenance remains useful structured evidence about how persistent memory was created. Any policy that turns this evidence into trust or reuse decisions still requires a separate downstream validation.

Reproducibility Statement

The appendix records the frozen intervention rules and statistical gates, together with the support-failure boundaries used by the controlled experiments. The submission supplement contains the claim ledger and its content-addressed evidence manifest. It also packages experiment receipts with the source needed to rebuild the manuscript and recompute the reported aggregate statistics.

AI Use Disclosure

LLM-based tools assisted literature triage and earlier hypothesis formulation. They supported experimental-methodology design under frozen claim contracts and helped implement experiment code. They also assisted result interpretation and manuscript editing. Separate LLM reviewers were used for mock-review analysis. Scientific claims and reported numerical results were checked against frozen experiment artifacts and deterministic evaluation scripts; AI-generated judgments were not used as task-success ground truth. The authors reviewed the AI-assisted work and take responsibility for the final content.

References

Marc-Antoine Allard, Arnaud Teinturier, Victor Xing, and Gautier Viaud. Experiential reflective learning for self-improving llm agents. In *ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems*, 2026.

Mohammad Asadolahi, Amir Amini, Samira Talebi, Amirfarhad Farhadi, and Azadeh Zamanifar. Memory reward inflation in self-improving llm agents, 2026.

Jialong Li and Jialing Zhu. Auditing self-evolution in financial agents: Capability gains, security drift, and execution-interface mismatch, 2026.

Yuxuan Liu, Zhaochen Su, Yuhao Zhang, et al. Rethinking self-evolving agent skills: Feedback dynamics over multiple rounds, 2026.

Siru Ouyang, Jun Yan, I-Hung Hsu, Yanfei Chen, Ke Jiang, Zifeng Wang, Rujun Han, Long T. Le, Samira Daruki, Xiangru Tang, Vishy Tirumalashetty, George Lee, Mahsan Rofouei, Hangfei Lin, Jiawei Han, Chen-Yu Lee, and Tomas Pfister. Reasoningbank: Scaling agent self-evolving with reasoning memory. In *International Conference on Learning Representations*, 2026.

Walid Saidi. Mutmem: Cryptographically authorized mutation in persistent agent memory, 2026.

Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, 2023.

Xueqiao Sun, Xiaohan Wang, Ludwig Schmidt, et al. Learning from failure: Inference-time self-improvement for computer-use agents, 2026.

Yiqi Wang, Jiaqi Zhang, Taotao Cai, et al. From agent traces to trust: A survey of evidence tracing and execution provenance in llm agents, 2026.

Zora Zhiruo Wang, Jiayuan Mao, Daniel Fried, and Graham Neubig. Agent workflow memory, 2024.

Jinghan Xu, Yiyong Xiao, Wanru Shao, et al. Memory provenance laundering in llm agents: A non-amplification firewall for persistent memory, 2026.

Wujiang Xu, Zujie Liang, Kai Mei, et al. A-mem: Agentic memory for llm agents, 2025.

Shengtao Zhang, Jiaqian Wang, Ruiwen Zhou, et al. Memrl: Self-evolving agents via runtime reinforcement learning on episodic memory, 2026.

A Source-Grounded Reanalysis and Intervention Details

A.1 Count reconstruction

The financial-agent audit reports 101 success-derived retrievals at 0.931 accuracy. The unique integer numerator consistent with this value at three decimal places is 94. It reports 34 failure-derived retrievals at 0.647 accuracy. The corresponding integer numerator is 22. This yields 7 failures in the success-derived stratum and 12 failures in the failure-derived stratum.

The same audit reports ten W→W persistent failures, all with failure-derived retrieval. Hence 10 of the 12 failures in the failure-derived stratum are persistent failures. The remaining two failure-derived errors can contribute to the reported C→W regressions. The success-derived stratum contains no W→W case and contributes seven failures to the non-persistent error pool.

A.2 TERMINAL INFORMATION-EQUIVALENCE CONTRACT

Six terminal candidates are frozen before downstream outcomes. Both success-derived and failure-derived memories are written with the same writer model. A candidate is eligible only when both outputs parse as valid ReasoningBank memory items. The cosine between paired memories must be at least 0.75. An independent verifier must mark their actionable guidance equivalent and must find no unique task-solving fact. Literal benchmark-reference leakage excludes a pair before downstream evaluation.

Four candidates satisfy every rule. Their released task-ID set is $\{387, 167, 23, 388\}$. Their embedding cosines are 0.931, 0.902, 0.906, and 0.904. Candidate 385 is excluded because a memory repeats the reference name "Cally." Candidate 164 is excluded because its memories repeat the references "Dry" and "Uneven color." These exclusions occur before downstream outcome generation.

A.3 TERMINAL RUNTIME REPAIR BOUNDARY

The condition-blind evidence projection removes repeated global menu blocks. It drops blank lines. It also removes exact duplicate lines. It uses neither benchmark references nor generated outcomes. The first downstream attempt is cancelled after a context-allocation support failure. No partial result is admitted to the scientific dataset. The clean restart changes only the allocated context window. The largest frozen prompt occupies 5,209 input tokens inside a 12,288-token allocation. The executor and sampling temperature remain frozen. The seed schedule and pair set are unchanged. The scorer is fixed. The effect floor is fixed. The permutation seed and significance threshold are fixed.

A.4 TERMINAL STATISTICAL RULE

Each qualified terminal pair is evaluated under six paired seeds. The primary statistic is the mean terminal-success difference between success-derived and failure-derived memory. Support requires a difference of at least 0.15 and a one-sided stratified permutation value below 0.05. Counterevidence requires a difference at or below -0.15 with the corresponding one-sided value below 0.05. The test uses 100,000 permutations with a frozen random seed. The observed difference is +0.1667 and the support-side value is $p = 0.0785$. Because the independent unit is the task pair, four units yield only 16 exact sign assignments. The exact one-sided resolution floor is $1/16 = 0.0625$, making this endpoint non-decisive under the frozen 0.05 rule.

A.5 EARLY-ACTION COHORT AND PROVENANCE INTERVENTION

The early-action endpoint uses a disjoint released WebArena cohort. All D2 task IDs used by earlier bridge experiments are excluded before selection. A candidate must come from a released successful trajectory and expose an intermediate browser state with a visible single-click reference action. The earliest qualifying step at index two or later is selected. Eligible tasks are sorted by numeric task ID and the first 24 are frozen before counterfactual writer outputs.

For each task, a compact process trace preserves the task instruction and executed action sequence. The serializer removes the terminal success Boolean. The same trace is then passed through the first-party ReasoningBank success-reflection prompt and failure-reflection prompt. Qwen2.5-32B is the writer at temperature zero. This creates a direct intervention on provenance mode while holding the process trace fixed.

The information-equivalence gate requires valid memory-item parsing and embedding cosine of at least 0.75. An independent verifier must find equivalent actionable guidance. It must also find no unique task-solving fact and no direct leakage of the frozen reference next action. Twenty-three pairs pass. Candidate 159 is excluded because the verifier output is unparsable. The downstream set is frozen as all 23 eligible pairs.

A.6 EARLY-ACTION STATISTICAL RULE

Each early-action task receives six paired seeds under each provenance condition. The executor sees the same future query and the same frozen browser state within a pair. Its required output is one visible click index. The released successful trajectory supplies the reference click.

The independent statistic is computed at task level. For each task, we subtract the failure-derived reference-match rate from the success-derived rate. The permutation test sign-flips these task-level differences over 100,000 draws. Support requires a mean difference of at least 0.15 and a one-sided value below 0.05. Counterevidence requires a mean at or below -0.15 with its one-sided value below 0.05.

Twenty-three tasks produce 276 completed requests with zero support failures. The mean reference-match difference is -0.0942. The support-side value is 0.9841. The counterevidence-side value is 0.0792. Paired actions differ in 21.0% of seeds. Hypothesis-direction error-consistent divergence occurs in 0.7% of seeds, while the reverse pattern occurs in 10.1%.

A.7 Exact-information support-capacity stop

A separate preregistered experiment holds the actionable memory core byte-identical and changes only structured provenance metadata. The frozen design requires ten unique review tasks under a fixed prompt-family contract. The entire released shopping split contains only five unique tasks under those rules. The experiment therefore stops before model execution. Its support-capacity failure cannot be converted into a scientific negative, and the target count is not changed after the audit.

A.8 Reopen condition

The strongest confirmation path is an exact financial AgentDojo matched-provenance replication when first-party query-level artifacts or auditable exact-model access become available. Any additional controlled bridge must be independently frozen before its outcomes and cannot reuse the R4 or R6 cohorts for threshold rescue.